Deep Learning Indaba 2026 · Skills Session

Doing ML Research *from the Margins*

A practical map for getting into serious ML research without institutional backing — and publishing at venues that matter.

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A 60-minute map

01

Framing

What the path looks like in retrospect

10 min

02

Finding mentors

Making contact that actually gets a response

15 min

03

Scoping research

Tractable problems without a GPU cluster

10 min

04

Reading the literature

From zero to frontier-aware, efficiently

10 min

05

Top venue positioning

Writing and framing work for where it belongs

10 min

Section 1 of 5

What the path looks like in retrospect

01

Luck isn't randomly distributed. The moves that break people through are identifiable.

The core premise

Luck in research is not randomly distributed.

"Cumulative advantage is the hypothesis that the more privilege you had in life, the more likely you had the resources (money, location, educated parents, mentoring, good peers, free time, extracurricular activities, extensive social network) to do well (rapid development, good grades), and this gives you more resources (better schools, better jobs, better connections) to do even better (promotions, tenure, grants), which yields even more resources (even more extensive social network, collaborations, funding) to do even better (Nobel Prize, unicorn startups)."

The core premise

Luck in research is not randomly distributed.

The people who break through from constrained contexts tend to make a small number of identifiable moves. This session makes them visible upfront.

PEOPLE WHO WALKED THIS PATH

Different starting points, but same navigable territory.

Steven K.

From: FUNAAB Undergrad, no lab

→ CMU PhD, science @FAANG, NeurIPS

Ifihan O.

From: Practitioner, Wits University

→ ML Collective Africa researcher

Chisom

From: Nigerian undergrad, MLC member

→ Cambridge MSc

Zaynab A.

From: UNILAG Undergrad

→ Max Planck Institute

Oreva A.

From: AI Saturdays Lagos

→ Google, top NLP publications

Yusuf

From: UNILAG undergrad

→ CMU Africa / NTU Singapore

Four moves, every trajectory

01

Find community early

Open-science spaces — ML Collective, IndabaX, AI Saturdays — give access to peers and senior researchers before you have any credentials.

02

Make yourself a worthwhile bet

Do a small piece of work first. Cold outreach with something to show converts at a completely different rate than cold outreach with nothing.

03

Scope first work carefully

Your first paper doesn't need to be ambitious. It needs to be done. A completed workshop paper beats an unfinished full paper every time.

04

Build momentum from small wins

Workshop → venue feedback → stronger submission. Each step makes the next one easier and builds the track record that opens doors.

Section 2 of 5

Finding and approaching mentors

02

Where to find accessible researchers, what gets a response, and how to build the relationship over time.

The mentor-finding problem is a funnel

STAGE 1

Find the right people

Researchers working on tractable problems who are actually reachable. Not necessarily the most famous — the most accessible.

STAGE 2

Make contact that gets a response

Most outreach fails here. The difference is specificity, not sincerity. Generic emails are skipped regardless of how earnest.

← most outreach fails here

STAGE 3

Build the relationship

Convert a response into an ongoing working relationship. Show momentum. Update regularly. Don't disappear for months.

Where to find accessible senior researchers

ML Collective & open-science communities

Researchers here are explicitly accessible and often looking for motivated collaborators. This is the warmest cold outreach possible.

Workshop author lists

Find people who just published at a relevant workshop. They're actively working in an adjacent area and recent submissions show momentum.

Paper acknowledgments & footnotes

'The authors thank X for helpful discussions' — X is often a PhD student or postdoc open to collaboration, not a busy PI.

X / Twitter engagement

Researchers who actively post and reply to questions are signalling accessibility. Engage publicly before reaching out.

IndabaX & local events

Researchers who attend African AI events are invested in the ecosystem. Meeting at an event changes the coldness of outreach.

Semantic Scholar / Google Scholar

Find mid-career researchers with 5–15 papers in your area. Senior enough to matter, junior enough to respond.

Anatomy of an email that lands

THE HOOK

Show you've read their work. Name one specific thing — a result, a technique, a framing. Generic compliments are invisible.

THE ASK

Small and concrete. A specific technical question, or a 15-minute call. Not 'be my mentor' or 'review my work.'

YOUR CONTEXT

Two or three sentences max. What you've built, what you're working toward. Not your life story.

THE EXIT RAMP

'No worries if you don't have bandwidth.' Gives them a graceful out — and makes you look confident, not desperate.

What not to do

- ✕ Generic email that could go to anyone
- ✕ Asking for mentorship upfront
- ✕ No concrete technical question
- ✕ Long email with your full life story
- ✕ Mass-emailing the same template

They talk to each other.

Resource: MLC Africa Mentor Outreach Templates → mlcollective.org/wiki/mlc-nigeria/mentor-outreach-email-templates

Timing and follow-up strategy

When to reach out

Best — Right after a workshop acceptance — you have momentum and something to show

Good — After meeting them at a conference or workshop

Okay — When you have a concrete, well-formed technical question

Bad — Before you have any work done, or when asking them to generate ideas for you

After you get a response

→ Update every 2–3 weeks with concrete progress

→ Keep it short — a paragraph, not an essay

→ Show momentum: what you ran, what you found, what's next

→ Don't disappear for months then reappear asking for a rec letter

→ One polite follow-up if no reply after a week. Then move on.

Templates for every scenario → mlcollective.org/wiki/mlc-nigeria/mentor-outreach-email-templates

Exercise

Draft your outreach email — now.

Think of one researcher whose work is adjacent to something you're interested in.

Use the MLC Africa templates as your scaffold. You have 4 minutes.

Five templates available:

1. Initial contact after workshop paper
2. Warm follow-up
3. Industry researchers
4. After a conference meeting
5. Cold outreach (use sparingly)

mlcollective.org/wiki/mlc-nigeria/mentor-outreach-email-templates

Section 3 of 5

Scoping tractable research

03

Finding problems that are meaningful, publishable, and executable with what you have.

The three-property tractability filter

A tractable problem for a resource-constrained researcher satisfies all three:

1 **No proprietary data or serious compute required for initial signal**

If you need a closed dataset or 8 A100s to run your first experiment, you've already lost. Find the version of the problem where open weights and a colab notebook get you 80% of the signal.

2 **Has a completable publishable unit**

Your first paper doesn't need to be ambitious — it needs to be done. Define the smallest version of the contribution that is honest and novel. A tight workshop paper beats an unfinished full conference paper.

3 **Close enough to the frontier that top venues will care**

This is where most people start. The mistake is optimizing only for this while ignoring (1) and (2). All three have to be true at once.

Your local context is an asset, not a liability

The reframe

Problems that are invisible to well-resourced labs are often exactly where the gap is.

Top labs have no incentive to work on low-resource languages, African clinical datasets, or evaluation gaps in non-English contexts. That's not a weakness — that's your comparative advantage.

The frontier is not one place. It has gaps you can see that they can't.

Worked examples

Afri-MCQA (ACL 2026)

Multilingual QA benchmark for African languages. Invisible problem to labs with no African language speakers.

NLP for low-resource languages

Masakhane demonstrated that African researchers working on African problems can publish at ACL, EMNLP, and beyond.

Dataset collection at local institutions

Hand-collected sign language data at Nigerian special education schools → National AI Champion recognition.

Writing the paper IS the research

Don't wait until you have results *to start writing.*

Writing the paper forces you to be clear about what you don't understand.

It crystallises the actual contribution. It opens dialogue with others.

Writing is not how you report the research — it's how you do it.

Simon Peyton Jones · "How to Write a Great Research Paper" · simon.peytonjones.org/great-research-paper

Andrej Karpathy · "A Survival Guide to a PhD" (problem selection section) · karpathy.github.io/2016/09/07/phd

Section 4 of 5

Reading the literature efficiently

04

From zero familiarity to frontier-aware in a week — a practical workflow.

The three-pass method

Pass 1

5–10 minutes

Bird's-eye view

- → Read title, abstract, introduction
- → Read section headings only
- → Read the conclusions
- → Glance at references

Output:

Can you name the category, context, contributions, and correctness? If not, stop here for papers outside your area.

Pass 2

Up to 1 hour

Content, not details

- → Read carefully, skipping proofs
- → Study all figures and graphs
- → Note relevant unread references
- → Jot down key points in margins

Output:

You should be able to summarise the paper's main thrust with supporting evidence to someone else.

Pass 3

4–5 hours

Full re-implementation

- → Virtually re-implement the paper
- → Challenge every assumption
- → Identify hidden failings
- → Jot ideas for future work

Output:

Reserved for papers central to your research. You should be able to reconstruct the whole structure from memory.

Keshav · "How to Read a Paper" (three-pass method + literature survey workflow) · web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf

How to go from zero to frontier-aware in a week

1

Find 3–5 recent papers

Use Google Scholar or Semantic Scholar. Search your topic with recent year filters. Do a pass-1 scan on each.

2

Trace shared citations

Read the related work sections. Papers cited by multiple of your initial 3–5 are the canonical papers.

3

Find the key researchers

Repeated author names in bibliographies = who the community considers foundational. Visit their pages to see where they've published recently.

4

Identify the top venues

Best researchers publish at top conferences. That tells you which venues matter for your area — and where to aim.

5

Iterate

If your initial papers all cite one key paper you missed, get it. Repeat until convergence. Usually 2–3 rounds.

Start with: Google Scholar · Semantic Scholar · Hugging Face Papers · X / Twitter (follow the authors)

Engaging publicly builds reputation — and it costs nothing

Comment on papers you read

Post a brief take on a paper you read this week. What was the key insight? What didn't they address? Even a short thread signals active engagement to the community.

Share reading notes publicly

Paper summaries, mind maps, annotated PDFs. Other researchers find these, reference them, and reach out. Networking without the awkward small talk.

Ask questions in the open

When you're confused about a technique or a result, ask publicly on X or in community Discords. Authors often respond. So do other readers who had the same question.

Engage with researchers you want to approach

Reply to their posts before you cold email them. A thoughtful comment means your email isn't cold anymore — they've already seen your name.

Section 5 of 5

Writing and positioning for top venues

05

Why people from our context default to lower venues, and what to do about it.

How peer review actually works

The idealized version

Scholars with leisure time, gathered in a quiet space, carefully pouring over the nuances of your work.

Each contribution evaluated on its merits. Every point considered thoroughly before a verdict is reached.

The reality

A large, crowded marketplace. Reviewers are time-constrained, reading 6–10 papers in a short window. They are not your opponents — but they are not your advocates either.

The first-pass test: can a reviewer reconstruct your contribution in five minutes?

If not, the paper is at risk regardless of the underlying work.

Parikh, Batra & Lee · "How we write rebuttals" · deviparikh.medium.com/how-we-write-rebuttals-dc84742fece1

Positioning and writing are different skills

The same work can be a workshop paper or an ICLR submission depending on how the contribution is scoped and communicated.

Workshop framing

"We explore whether technique X can be applied to low-resource language Y and present preliminary results."

Signals: exploratory, preliminary, not confident in contribution.

Result: reviewer reads it as 'interesting but not ready.' Gets workshopped. Never becomes a full paper.

Conference framing

"We demonstrate that technique X, applied to low-resource language Y, achieves Z improvement — the first result of this kind."

Signals: confident, specific, novel, measurable.

Same underlying work. Same experiments. Positioned as a concrete contribution rather than an exploration.

Why researchers from our context default to lower venues

It is rarely about the quality of the work. It is almost always about navigation.

Don't know which venues matter

Without labmates and advisors to calibrate against, it's hard to know the hierarchy of venues — or that there is one.

Underestimate the contribution

Work that is genuinely novel gets positioned as preliminary or exploratory because there's no one to push back and say 'this is actually good.'

No feedback loop before submission

Labs iterate. They workshop papers internally, get senior feedback, and revise before submission. Independent researchers often submit cold.

Risk aversion

A workshop rejection feels safer than a main conference rejection. But workshops rarely lead anywhere. The calculus doesn't hold.

OpenReview: your free advisor

Actual reviewer comments on accepted and rejected papers in your area — right now, at no cost. The most honest signal available.

How to use it

1. Go to openreview.net
2. Search a venue in your area (ICLR, NeurIPS, EMNLP, ACL)
3. Filter by year — look at recent accepted papers
4. Find a paper adjacent to what you're working on
5. Read the reviews — all of them, including the low scores
6. Read the author responses (rebuttals)
7. Notice the pattern: what do reviewers consistently reward?

What to look for

Clarity of contribution

Reviewers reject papers where they can't articulate the contribution after one pass. Read to see how accepted papers nail this in the abstract.

Expected baselines

Reviewers consistently flag missing baselines. See which comparisons are expected in your subfield.

Framing novelty

How do accepted papers distinguish themselves from prior work? The language patterns are learnable.

Rebuttal wins and losses

Some author responses flip decisions. Some don't. The difference is instructive.

OpenReview · openreview.net (search ICLR, NeurIPS, EMNLP, ACL by year to read live reviews)

What top venue reviewers actually look for

Novelty	Is there a genuinely new idea here? Not incremental tuning — something the field doesn't already have.
Significance	Does it matter? Will it change how people think about this problem or enable something not previously possible?
Soundness	Are the claims supported by the evidence? Are the baselines fair? Are the assumptions stated?
Clarity	Can a reviewer reconstruct the contribution in five minutes from the abstract and intro? If not, the paper is at risk regardless of the work underneath.
Related work	Does the paper know its neighbours? A missing relevant citation is a flag. A well-positioned related work section is a trust signal.

Colin Raffel · "Reviewing Criteria" · colinraffel.com/blog/reviewing-criteria.html

TOP VENUE POSITIONING

A note on rebuttals — most people from our context don't know these exist

What a rebuttal is:

After reviews come in, authors get a short window to respond before the final decision. Most conferences give you ~500 words. Use them. Many papers flip from reject to accept at this stage.

You have two audiences

The reviewers (clarify doubts, correct misunderstandings) and the Area Chair (who makes the final call). Write for both. The AC may not read the full reviews.

Lead with the direct answer

Quote the concern concisely, then answer it immediately. Don't bury the response in context. Reviewers are skimming.

Pick your battles

Not every criticism needs a rebuttal. Focus on factual errors, misunderstandings about your method, and claims you can correct with new evidence.

Don't be defensive

The goal is to clarify and convince, not win an argument. Find points of agreement. It disarms hostile reviewers.

Parikh, Batra & Lee · "How we write rebuttals" · deviparikh.medium.com/how-we-write-rebuttals-dc84742fece1

The goal was never to inspire.

It was to leave you with a clearer map than we had.

The path exists. It's navigable. And now it's a little more legible.

All resources from this session

Section 2 — Finding mentors

MLC Africa Mentor Outreach Templates	mlcollective.org/wiki/mlc-nigeria/mentor-outreach-email-templates

Section 3 — Scoping research

Simon Peyton Jones — How to Write a Great Research Paper (video + slides)	simon.peytonjones.org/great-research-paper
Andrej Karpathy — A Survival Guide to a PhD	karpathy.github.io/2016/09/07/phd

Section 4 — Reading the literature

Keshav — How to Read a Paper (3-pass method)	web.stanford.edu/class/ee384m/Handouts/HowtoReadPaper.pdf

Section 5 — Top venue positioning

OpenReview — read actual reviewer comments on papers in your area	openreview.net
Colin Raffel — Reviewing Criteria	colinraffel.com/blog/reviewing-criteria.html
Parikh, Batra & Lee — How We Write Rebuttals	deviparikh.medium.com/how-we-write-rebuttals-dc84742fece1

Questions

& discussion

We're also around after the session.

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mlcollective.org/